Md. Shahriar Kabir

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Summary

Final-year Computer Science and Engineering student at the University of Dhaka and Backend AI Engineer Intern at FlyRank AI. Builds RAG systems, LLM tool-calling pipelines, and REST APIs with Python, FastAPI, and Node.js, with hands-on experience in retrieval evaluation, real-time web applications, and deploying machine learning models.

Experience

Backend AI Engineer Intern **Jun 2026 – Present**
FlyRank AI, Chicago, Illinois, USA *Remote*

- Design API contracts and retrieval-backed answer flows for AI product features, owning correctness and failure handling.
- Build structured-output and tool-calling workflows over LLM APIs, with schema validation and deterministic parsing of model responses.
- Implement RAG components (ingestion, chunking, embedding, retrieval) behind async FastAPI services.
- Write evaluation harnesses that score answer quality and catch regressions across prompt and retrieval changes; review pull requests and design docs with a distributed team.

Projects

Evidrix | Evidence-Grounded Question Answering GitHub
Python, FastAPI, PostgreSQL, pgvector, Redis, Docker

- Built a RAG system that answers questions over technical documentation and research papers, citing the exact passage and page behind every claim.
- Implemented hybrid retrieval (vector and full-text search) with cross-encoder reranking, improving MRR@5 by 0.193 over vector-only search on a held-out benchmark.
- Reached 86% answer correctness with fully faithful citations, and declined every question the documents could not answer.
- Hardened the system for production with per-user data isolation, API-key authentication, rate limiting, background ingestion with retries, and 469 automated tests (92% coverage).

MediSync | Healthcare Management Platform GitHub
React, Node.js, Express, PostgreSQL, Supabase, Socket.IO, WebRTC

- Built a multi-portal platform for doctors, patients, and diagnostic centers covering appointments, medical records, lab reports, and billing.
- Developed about 200 REST endpoints on a 46-table PostgreSQL schema.
- Implemented real-time chat with Socket.IO and in-browser video calls with WebRTC.
- Added an LLM-powered clinical assistant that summarizes each patient's lab history, JWT role-based access control, and online payments with automatic revenue sharing.

GridWise LLM | Energy Grid Optimization GitHub
Python, FastAPI, SciPy, Docker

- Built a FastAPI service that converts operators' plain-language notes into structured scheduling directives using LLMs.
- Optimized 24-hour grid, solar, and battery schedules for minimum cost with SciPy linear programming, guarded by rule-based safety checks.
- Added replay validation of every plan, automatic fallback between LLM providers, automated tests, and Docker packaging.

GrocyGenie | Grocery Consumption Prediction Live | GitHub
Python, scikit-learn, pandas, NumPy, FastAPI

- Trained a gradient-boosting model on a 270K-record synthetic dataset to predict daily consumption and run-out dates for 15 grocery products.
- Achieved R^2 0.989 and MAE 43.5 g/day on a held-out split of the synthetic dataset, with reproducible training and evaluation scripts.
- Deployed the model as a FastAPI service on Hugging Face Spaces, with a feedback loop that retrains the model from user corrections.

Digital Knowledge Platform | Academic Library System Live | GitHub
React, Node.js, Express, PostgreSQL, Supabase, Python, Docker

- Developed a digital library for the CSEDU department with document search, book circulation with automated fines, and a student project showcase.
- Built a Node.js REST API with JWT authentication and role-based access for admins, staff, and members.
- Created a Python microservice that classifies uploaded documents and extracts keywords and metadata; containerized the stack with Docker Compose.

FamNest | Family Finance and Organizer App GitHub

Flutter, Dart, FastAPI, MongoDB Atlas, Cloudinary

- Built an Android app for families to track shared expenses and budgets, plan events, and store important documents.
- Designed FastAPI REST endpoints over MongoDB Atlas and integrated Cloudinary for media storage.

Collaborative Text Editor | Real-Time Editing over LAN GitHub

Java, JavaFX, TCP sockets, multithreading

- Built a desktop editor where multiple users edit the same document in real time over a local network, with chat in every session.
- Designed a multithreaded TCP server with a thread pool configured for up to 100 clients and automatic cleanup of inactive sessions.

Disaster Management System | Disaster Response Coordination GitHub

Node.js, Express, MySQL, HTML, CSS

- Built a web app to coordinate disaster response across victims, volunteers, rescue operations, and donations.
- Designed the relational MySQL schema and REST CRUD APIs, secured with Helmet and rate limiting.

Education

University of Dhaka | B.Sc. in Computer Science and Engineering **Expected Dec 2026**

Dhaka, Bangladesh *CGPA: 3.60 / 4.00 (through 7th semester)*

Relevant coursework: Data Structures, Algorithm Analysis, Object-Oriented Programming, Database Management, Operating Systems, Computer Networks, Software Engineering, Machine Learning, Artificial Intelligence.
Research interests: Video anomaly detection, vision-language models, weakly supervised learning.

Skills

Programming Languages: Python, JavaScript, TypeScript, Java, C++, C, SQL, Dart
AI & Machine Learning: RAG, LLM APIs, tool calling, structured outputs, embeddings, hybrid search, reranking, LLM evaluation, scikit-learn, TensorFlow, pandas, NumPy, Hugging Face
Backend: FastAPI, Node.js, Express, REST APIs, Socket.IO, WebRTC, JWT authentication
Databases: PostgreSQL, pgvector, Supabase, Redis, MongoDB, MySQL, SQLite
Frontend & Mobile: React, Tailwind CSS, HTML, CSS, Flutter, JavaFX
DevOps & Tools: Docker, Docker Compose, Git, GitHub Actions, CI/CD, Linux

Achievements & Certifications

- **Competitive programming:** 400+ problems solved on CodeChef; Silver badges in Problem Solving and DSA on HackerRank.
- **Certifications:** Learn Python Pro, CodeChef (2023) | SQL (Basic), HackerRank (2025).
- **Languages:** Bangla (native), English (professional working proficiency).